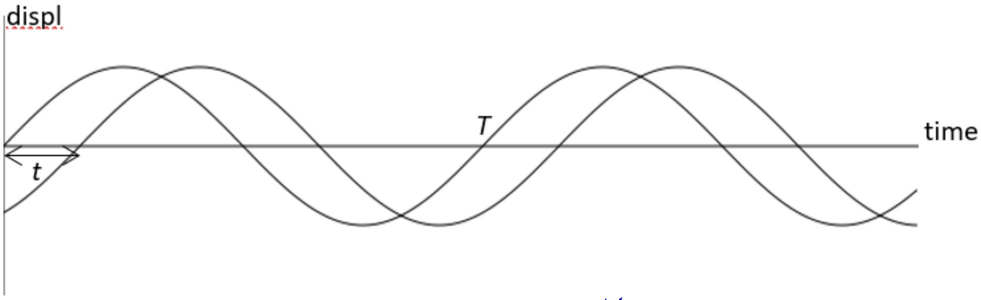
4 **(a)** Two waves are of the same frequency.

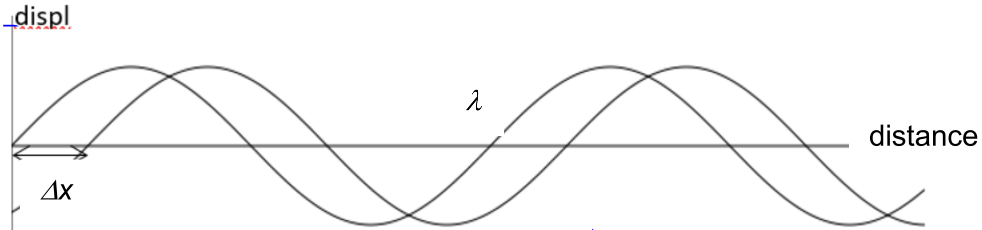
Explain with the aid of a diagram what, for the two waves, is meant by *phase difference*.



[1]

Phase difference between two waves is $\Delta\phi = \frac{\Delta t}{T} 2\pi$, where Δt is the time between two corresponding points, e.g. crest and crest or trough and trough on the two waves, and T is the period of the wave. [1]

[2]



Phase difference between two waves is $\Delta\phi = \frac{\Delta x}{\lambda} 2\pi$, where Δx is the distance between two corresponding points, e.g. crest and crest or trough and trough on the two waves, and λ is the wavelength of the wave. [1]

[1]

(b) Monochromatic light is incident normally on a double slit as shown in Fig. 4.1. Light passes through the two slits B and C and is incident on the screen.

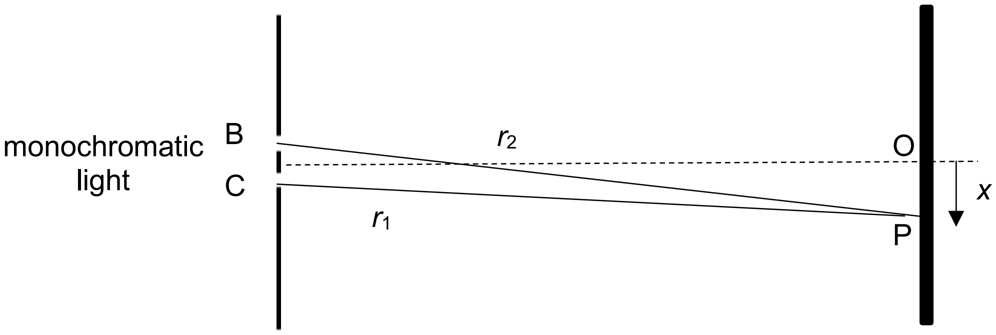


Fig. 4.1

The centre of the interference pattern formed on the screen is at O. The separation between the fringes is y.

r_1 and r_2 are two waves arriving at P.

(i) **1.** Deduce the relationship between the phase difference of the two waves arriving at point P and the distance x from point O.

$$\frac{\Delta\phi}{2\pi} = \frac{x}{y}$$

$$\Delta\phi = \frac{2\pi x}{y}$$

[1]

2. The waves have a phase difference of 12.6 radians when they meet at point P. Distance OP on the screen is 5.2 mm. Calculate the separation y between the fringes.

$$\Delta\phi = \frac{2\pi x}{y}$$

$$y = \frac{2\pi x}{\Delta\phi}$$

$$= \frac{2\pi \times 5.2}{12.6} \qquad [1]$$

$$y = 2.6 \text{ mm} \qquad [1]$$

y = ... m [2]

(ii) The light is adjusted so that the intensity of the light passing through slit B is reduced to a quarter that through slit C. The intensity of light from slit C alone at O is I. Deduce in terms of I, the intensity of the light at O due to the two slits.

Intensity $I = kA^2$ where k is a constant.

The amplitude of light from B is then $\frac{A}{2}$ [1]

The total intensity at O is thus $k\left(\frac{A}{2}+A\right)^2 = 2.25kA^2 = 2.25I$ [1]

intensity = ... [2]

(iii) Sketch, on Fig. 4.2, a graph to show the variation with distance x from point O of the intensity of light observed on the screen. Label your answer to **(b)(ii)** on Fig. 4.2. Ignore the single slit diffraction envelope.

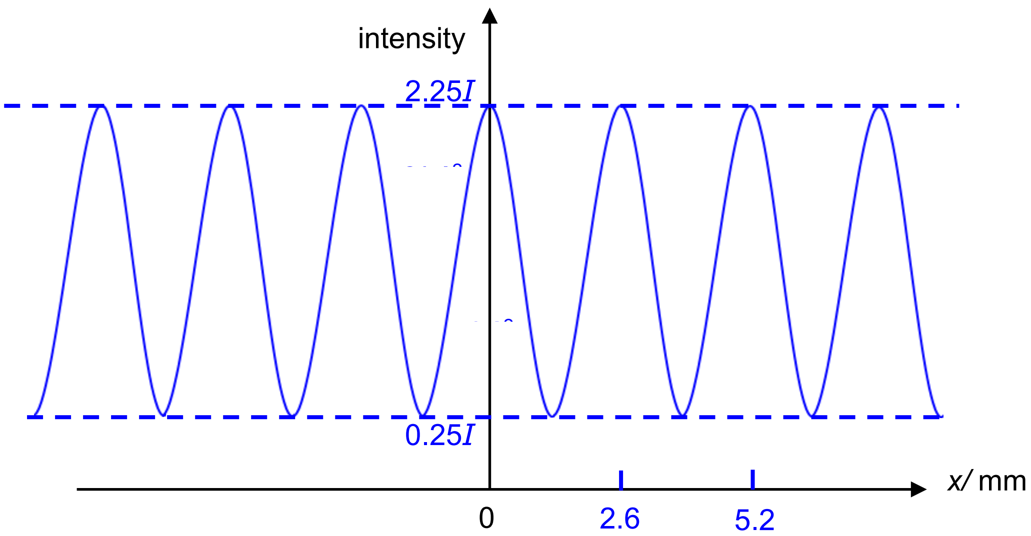


Fig. 4.2

[1m for max and min intensities at $2.25I$ and $0.25I$ respectively. 1m for constant fringe separation]

[2]

[Total: 9]